

Relational Model

COSC 304 – Introduction to Database Systems



Relational Model History

The relational model was proposed by E. F. Codd in 1970. Commercial implementations appeared in the late 1970s and early 1980s.

One of the first relational database systems, System R, developed at IBM, led to several important breakthroughs:

- development of SEQUEL, which evolved into SQL
- influence on commercial relational systems such as Oracle and IBM DB2
- extensive research on concurrency control, transaction management, and query processing and optimization

The relational model remains the foundation of many of the most widely used database systems, although relational systems increasingly coexist with document, graph, and other data models.



The Relational Model: Terminology

The **relational model** organizes data into tables called relations.

A **relation** is a table with columns and rows.

An **attribute** is a named column of a relation.

A **tuple** is a row of a relation.

A **domain** is a set of allowable values for an attribute.

The **degree** of a relation is the number of attributes it contains.

The **cardinality** of a relation is the number of tuples it contains.

The **intension** is the structure (schema) of the relation, including its attributes and domains.

The **extension** is the set of tuples currently in the relation.

Relation Example

relation Emp attributes							
	eno	ename	bdate	title	salary	supereno	dno
+	E1	J. Doe	1/5/1975	EE	\$30,000.00	E2	
+	E2	M. Smith	6/4/1966	SA	\$50,000.00	E5	D3
+	E3	A. Lee	7/5/1966	ME	\$40,000.00	E7	D2
+	E4	J. Miller	9/1/1950	PR	\$20,000.00	E6	D3
+	E5	B. Casey	12/25/1971	SA	\$50,000.00	E8	D3
+	E6	L. Chu	11/30/1965	EE	\$30,000.00	E7	D2
+	E7	R. Davis	9/8/1977	ME	\$40,000.00	E8	D1
+	E8	J. Jones	10/11/1972	SA	\$50,000.00		D1
*					\$0.00		

Record: 1 of 8

No Filter

Search

Degree = 7
Cardinality = 8

Domain of salary is set of
non-negative monetary values

Relational Model Formal Definition

The relational model is formally defined in terms of sets and set operations.

A **relation schema** $R(A_1, A_2, \dots, A_n)$ consists of attributes A_i , where each attribute has a name and a domain $dom(A_i)$.

A **relation instance** denoted $r(R)$ is a set of n -tuples $\langle d_1, d_2, \dots, d_n \rangle$ where each d_i is an element of $dom(A_i)$ or is `NULL`.

- The relation instance is the *extension* of the relation.
- A `NULL` value represents a value that is unknown or not applicable.

Example: Product (id, name, supplierId, categoryId, price)

- R = Product (relation name)
- Set $A = \{id, name, supplierId, categoryId, price\}$
- $dom(price)$ is set of all allowable non-negative monetary values
- $dom(name)$ is set of all possible strings that represent product names

Relation Practice Questions

eno	pno	resp	hours
E1	P1	Manager	12
E2	P1	Analyst	24
E2	P2	Analyst	6
E3	P3	Consultant	10
E3	P4	Engineer	48
E4	P2	Programmer	18
E5	P2	Manager	24
E6	P4	Manager	48
E7	P3	Engineer	36

Record: 1 of 9 | No Filter | Search

- 1) What is the name of the relation?
- 2) What is the cardinality of the relation?
- 3) What is the degree of the relation?
- 4) What is the domain of `resp`? What is the domain of `hours`?
- 5) Is the extension (instance) always larger than the intension (schema)?
- 6) Is a relation's cardinality always bigger than its degree?

Database Definition Matching Question

Question: Given the three definitions, select the ordering that contains their related definitions.

1) relation

2) tuple

3) attribute

A) column, row, table

B) row, column, table

C) table, row, column

D) table, column, row

Cardinality and Degree Question

Question: A database table has 5 rows and 10 columns. Select **one** true statement.

- A)** The table's degree is 50.
- B)** The table's cardinality is 5.
- C)** The table's degree is 5.
- D)** The table's cardinality is 10.

Relation Properties

- 1) Each relation has a unique name within a database schema.
- 2) Each attribute within a relation has a unique name.
- 3) Each tuple is distinct; a relation contains no duplicate tuples.
- 4) The order of attributes has no semantic significance.
- 5) The order of tuples has no significance.

Note: SQL tables and query results may contain duplicate rows; the pure relational model does not.



Relational Keys

Keys are used to uniquely identify a tuple in a relation.

- Keys apply to the schema not to the data. That is, looking at the current data cannot tell you for sure if the set of attributes is a key.

A **superkey** is a set of attributes that uniquely identifies a tuple in a relation.

A (**candidate**) **key** is a *minimal* set of attributes that uniquely identifies a tuple in a relation.

- A relation may have multiple candidate keys, and candidate keys may contain different numbers of attributes.

A **primary key** is the candidate key selected as the primary identifier for tuples in a relation.

A **foreign key** is a set of attributes in one relation that refers to a key in the same or another relation.

- Foreign keys enforce referential integrity.

Keys and Superkeys Question

Question: True or false: A key is always a superkey.

A) true

B) false

Keys and Candidate Keys Question

Question: True or false: It is possible to have more than one key for a table and the keys may have different numbers of attributes.

A) true

B) false

Keys and Data Question

Question: True or false: It is possible to always determine whether a set of attributes is a key by looking at the data in the relation.

A) true

B) false

Example Relational Data Questions

Keys are underlined

Emp Relation

<u>eno</u>	ename	title	salary
E1	J. Doe	EE	30000
E2	M. Smith	SA	50000
E3	A. Lee	ME	40000
E4	J. Miller	PR	20000
E5	B. Casey	SA	50000
E6	L. Chu	EE	30000
E7	R. Davis	ME	40000
E8	J. Jones	SA	50000

WorksOn Relation

<u>eno</u>	<u>pno</u>	resp	hours
E1	P1	Manager	12
E2	P1	Analyst	24
E2	P2	Analyst	6
E3	P3	Consultant	10
E3	P4	Engineer	48
E4	P2	Programmer	18
E5	P2	Manager	24
E6	P4	Manager	48
E7	P3	Engineer	36

Proj Relation

<u>pno</u>	pname	budget
P1	Instruments	150000
P2	DB Develop	135000
P3	Budget	250000
P4	Maintenance	310000
P5	CAD/CAM	500000

Questions:

- 1) Is *ename* a key for *emp*?
- 2) Is *eno* (by itself) a key for *WorksOn*?
- 3) List all the superkeys for *WorksOn*.

Keys Practice Question

Consider a relation storing driver information:

Driver Relation

SIN	name	plateNum	province
700 000 001	Alex Chen	ABC123	BC
600 000 002	Maya Singh	ABC123	AB
700 000 004	Jordan Lee	KLM742	BC
400 000 004	Alex Chen	ZYX908	ON

Questions:

- 1) List the candidate keys for the relation.
- 2) Pick a primary key for the relation.
- 3) Is *name* a candidate key?
- 4) List all the superkeys for the relation.

Assumptions:

- 1) Each driver has exactly one licence plate.
- 2) Each (plateNum, province) is associated with exactly one driver.
- 3) A plate number is unique within a province.



Relational Integrity

Integrity constraints are rules that restrict the values and relationships that may be stored in order to maintain valid and consistent data.

Types of constraints:

- **Domain constraint** - Every value for an attribute must be an element of the attribute's domain or be NULL.
 - NULL represents a value that is currently unknown or not applicable.
 - NULL is not the same as zero or an empty string.
- **Entity integrity constraint** - No attribute of a primary key can be NULL.
- **Referential integrity constraint** - A non-NULL foreign key value must match an existing value of the referenced key.

Foreign Keys Example

Emp Relation

<u>eno</u>	ename	title	salary
E1	J. Doe	EE	30000
E2	M. Smith	SA	50000
E3	A. Lee	ME	40000
E4	J. Miller	PR	20000
E5	B. Casey	SA	50000
E6	L. Chu	EE	30000
E7	R. Davis	ME	40000
E8	J. Jones	SA	50000

WorksOn.eno is
FK to Emp.eno

WorksOn.pno is
FK to Proj.pno

WorksOn Relation

<u>eno</u>	<u>pno</u>	resp	hours
E1	P1	Manager	12
E2	P1	Analyst	24
E2	P2	Analyst	6
E3	P3	Consultant	10
E3	P4	Engineer	48
E4	P2	Programmer	18
E5	P2	Manager	24
E6	P4	Manager	48
E7	P3	Engineer	36

Proj Relation

<u>pno</u>	pname	budget
P1	Instruments	150000
P2	DB Develop	135000
P3	Budget	250000
P4	Maintenance	310000
P5	CAD/CAM	500000

Foreign Keys Need Not Be Primary Keys

Proj Relation

<u>pno</u>	pname	budget	dno
P1	Instruments	150000	D1
P2	DB Develop	135000	D2
P3	Budget	250000	D3
P4	Maintenance	310000	null
P5	CAD/CAM	500000	D1

Proj.dno is
FK to Dept.dno

Dept Relation

<u>dno</u>	dname
D1	Management
D2	Consulting
D3	Accounting
D4	Development

An attribute can be a foreign key without being part of that relation's primary key.

Integrity Constraints Question

Question: What constraint says that a primary key attribute cannot be NULL?

- A) domain constraint
- B) referential integrity constraint
- C) entity integrity constraint

Entity Integrity Constraint Question

Question: A primary key has three attributes. Only one attribute is NULL. Is the entity integrity constraint violated?

A) Yes

B) No

Referential Integrity Constraint Question

Question: A foreign key has a NULL value.

Is the referential integrity constraint violated?

A) Yes

B) No

Integrity Questions

Emp Relation

<u>eno</u>	ename	title	salary
E1	J. Doe	EE	AS
E2	null	SA	50000
E3	A. Lee	null	40000
E4	J. Miller	PR	20000
E5	B. Casey	SA	50000
null	L. Chu	EE	30000
E7	R. Davis	ME	null
E8	J. Jones	SA	50000

Proj Relation

<u>pno</u>	pname	budget
P1	Instruments	150000
P2	DB Develop	135000
P3	Budget	250000
P4	Maintenance	310000
P5	null	null

WorksOn Relation

<u>eno</u>	<u>pno</u>	resp	hours
E1	P0	null	12
E2	P1	Analyst	null
null	P2	Analyst	6
E3	P3	Consultant	10
E9	P4	Engineer	48
E4	P2	Programmer	18
E5	null	Manager	24
E6	P4	Manager	48
E7	P6	Engineer	36
E7	P4	Engineer	23
null	null	Manager	40

Question: How many rows violate at least one integrity constraint? Note: salary, budget, hours have number domains.

- A) 8 B) 9 C) 10 D) 11 E) 12

Conclusion

The **relational model** represents data as relations which are sets of tuples. Each relational schema is a set of attributes with domains.

- A relation is a table with columns and rows. An attribute is a named column. A tuple is a row.
- Degree is the number of attributes, and cardinality is the number of tuples.

Keys are used to uniquely identify tuples in relations.

- A superkey is any set of attributes that uniquely identifies a tuple in a relation.
- A candidate key is a minimal superkey.

The relational model provides **constraints** to enforce data integrity, including domain, entity integrity and referential integrity constraints.

Objectives

- Define: relation, attribute, tuple, domain, degree, cardinality, intension, extension, relation schema, relation instance, NULL
- List the properties of relations.
- Define: superkey, key, candidate key, primary key, foreign key
- Define and distinguish domain, entity integrity, and referential integrity constraints.
- Given a relation, be able to:
 - identify its cardinality, degree, domains, candidate keys, and superkeys
 - determine whether integrity constraints are violated



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